Polynomial Functions

A **polynomial function** in standard form is

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \qquad a_n \neq 0,$$

where the coefficients are real numbers and the exponents are nonnegative integers.

Degree	n, the largest exponent with a nonzero coefficient
Leading term	$a_n x^n$
Leading coefficient	a_n
Constant term	$a_0 = f(0)$

Every polynomial has domain $(-\infty, \infty)$

- $f(x) = mx + b$, $f(x) = ax^2 + bx + c$

Ex: $f(x) = x^3 + 100x - 710$, $f(x) = 100x^{60} - x^2 + 1$

degree 3 degree 60

Recognize a Polynomial

Example. Which functions are polynomials? Explain any exclusions. For each polynomial, write standard form and identify its degree, leading coefficient, and constant term.

(a) $p(x) = 7 - 3x + 2x^4$ (b) $q(x) = \sqrt{x} + 1$ (c) $r(x) = \dfrac{x^2 - 4}{x - 2}$ (d) $s(x) = 8$

✓

in standard form.

Degree: 4

lead. term: $2x^4$

leading coef: 2

$q(x) = x^{\frac{1}{2}} + 1$

$r(x) = \dfrac{x^2 - 2^2}{x - 2}$

$= \dfrac{(x+2)(x-2)}{x-2}$

$= x + 2$

constant polynomial

degree is 0,

leadi term is 8

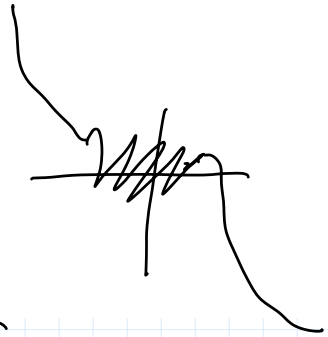
... coe is 8

constant is 8

Leading Term and End Behavior

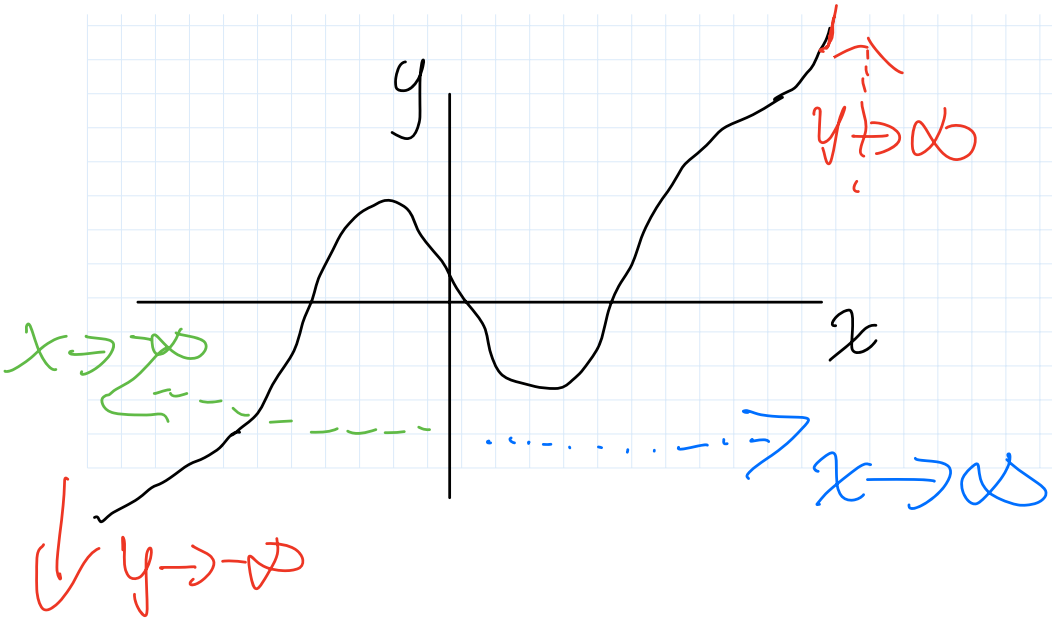
For large $|x|$, a polynomial has the same end behavior as its leading term $a_n x^n$.

Degree	Leading coefficient	As $x \to -\infty$	As $x \to \infty$
Even	$a_n > 0$	$f(x) \to \infty$	$f(x) \to \infty$
Even	$a_n < 0$	$f(x) \to -\infty$	$f(x) \to -\infty$
Odd	$a_n > 0$	$f(x) \to -\infty$	$f(x) \to \infty$
Odd	$a_n < 0$	$f(x) \to \infty$	$f(x) \to -\infty$



Example. For $p(x) = -2x^5 + 3x^2 - 7$, identify the leading term and describe both ends. Complete the limits.

$\lim_{x \to -\infty} p(x) =$ ∞ $\lim_{x \to \infty} p(x) =$ $-\infty$



- $-2x^5$

degree = 5, odd

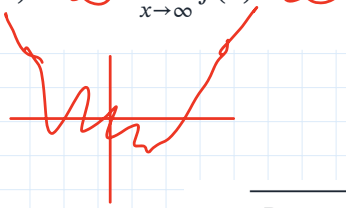
leading coe. = $-2 < 0$

Write the Limits

Example. Use the leading term to determine both limits for each polynomial. Sketch arrows showing its left and right ends.

(a) $f(x) = 3x^4 - 5x + 2$

$\lim_{x \to -\infty} f(x) = \infty$ $\lim_{x \to \infty} f(x) = \infty$



$3x^4$

degree $= 4$, even

lead. coe $= 3$, > 0

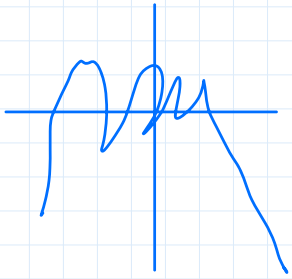
Degree	Leading coefficient	As $x \to -\infty$	As $x \to \infty$
Even	$a_n > 0$	$f(x) \to \infty$	$f(x) \to \infty$
Even	$a_n < 0$	$f(x) \to -\infty$	$f(x) \to -\infty$
Odd	$a_n > 0$	$f(x) \to -\infty$	$f(x) \to \infty$
Odd	$a_n < 0$	$f(x) \to \infty$	$f(x) \to -\infty$

(b) $g(x) = -x^6 + 4x^2$

$\lim_{x \to -\infty} g(x) = -\infty$ $\lim_{x \to \infty} g(x) = -\infty$

deg. $= 6$, even

$-1, < 0$



Zeros and Multiplicity

A real number r is a **zero** if $f(r) = 0$; the corresponding intercept is $(r, 0)$.

A factor $(x - r)^m$ gives zero r with **multiplicity** m.

Odd multiplicity: the graph **crosses** the x-axis. Even multiplicity: it **touches and turns**.

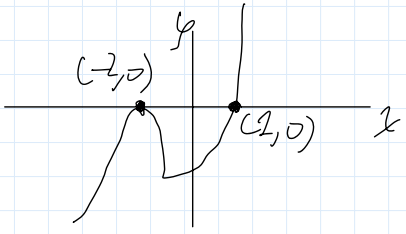
Example. Find the zeros of $f(x) = (x+2)^2(x-1)^3$. Give each multiplicity and state whether the graph crosses or touches. Find the degree and leading coefficient.

$f(x) = (x+1)^2(x-1)^3 = 0$

$= (x+2)\cdot(x+2)\cdot(x-1)\cdot(x-1)\cdot(x-1)$

$= 1x^5 + (x^4 + \dots + 0$

degree is 5, leading coeff. is 1.



$\sqrt[2]{(x+2)^2} = \sqrt[2]{0}$

$x+2 = 0$

$x = -2$

with multiplicity 2

$(x-1)^3 = 0$

$x-1 = 0$

$x = 1$

with multiplicity 3

Zero	x-int	Multip	behavior
-2	(-2,0)	even	Touches and turn
1	(1,0)	odd	Crosses

$$P(x) = -(x+2)(x-1)^2$$

$$= -(2x+2)(2x-1)(2x-2)$$

$$-2x^3$$

Turning Points

A degree-n polynomial has **at most** $n-1$ **turning points**.

If a polynomial graph has k turning points, its degree is **at least** $k+1$.

(a) For $f(x) = -2(x+1)^2(x-3)$, find the degree and the maximum possible number of turning points.

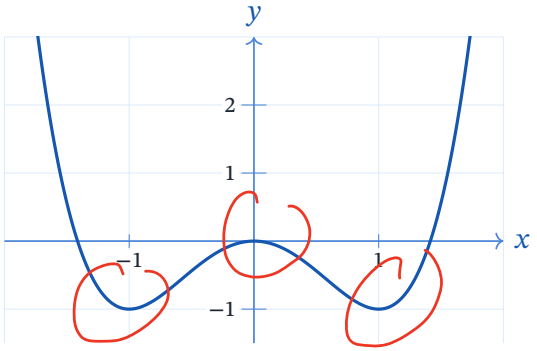
$(x+1)^2 \to x^2$

$(x-3) \to x$ $\Rightarrow x^3$

degree is 3

at most 2 turning.

(b) Count the turning points in this polynomial graph. What is its minimum possible degree?



degree is at least 4.

Put the Information Together

Example. Let $p(x) = -2(x+3)(x-1)^2$.

(a) Find the degree, leading term, and y-intercept.

(b) Find the zeros and multiplicities. State crossing or touching at each zero.

(c) Write both end-behavior limits and give the maximum possible number of turning points.

as degree 3, $-2x^3$, $P(0) = -2(0+3)\cdot(0-2)^2$

$= -2\cdot 3\cdot(-2)^2 = -6$, $(0,-6)$

(b) $-2(x+3)\cdot(x-1)^2 = 0$

$(x+3)=0$, $(x-1)^2=0$

$x=-3$, $x=1$, multiplicity 2

(c) max 2 turning point

odd degree, negative leading coefficient

$\begin{cases} \lim_{x\to\infty} P(x) = \infty \\ \lim_{x\to\infty} P(x) = -\infty \end{cases}$